1. Views

- A saved question with a name
- Use it like a table
- Always fresh: no copy of the data

```
SELECT * FROM open_loans;
```

loan_id	name	title
2	Aisha	Wonder
5	Carlos	Harry Potter
7	Ben	Dog Man
8	Dana	Holes

2. Indexes

- Like the index at the back of a book
- CREATE INDEX name ON table(column)
- Great for WHERE, JOIN and ORDER BY columns

```
EXPLAIN QUERY PLAN
SELECT * FROM books
WHERE author = 'Roald Dahl';
```

id	parent	notused	detail
3	0	0	SEARCH books USING INDEX idx_books_author (author=?)

SEARCH ... USING INDEX = fast!

3. Transactions

- BEGIN starts, COMMIT saves everything
- All or nothing
- ROLLBACK = the undo button

4. Constraints recap

- PRIMARY KEY, FOREIGN KEY
- NOT NULL, UNIQUE
- CHECK (a test) and DEFAULT (a fallback value)

5. Practice free + what's next

- sqliteonline.com, db-fiddle.com, sqlime.org (choose SQLite)
- sqlbolt.com for free lessons
- Paste the setup from the practice guide, click Run
- Next, Part 9: database design